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Financial concerns: a critical gap in survivors' experience of the implementation of the national standards for cancer survivorship care

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Abstract

Purpose It is vitally important to comprehensively identify and address cancer survivors' unmet needs. The new national standards for cancer survivorship care may improve survivorship care by defining quality care. The purpose of this convergent mixed-methods study was to describe how survivorship care at one cancer center aligns with the health system processes outlined in the national standards.

Methods Survey participants (N = 150) were survivors purposively sampled by treatment status, cancer stage, and geographic residence. Survey data evaluated how well providers assessed, listened to, and managed participants' concerns across key domains and were primarily analyzed descriptively. Survey data were complemented by semi-structured interview data from n = 17 survivors, which were analyzed using directed content analysis.

Results Greater than 70% of participants reported that their providers regularly assessed and listened to physical and emotional concerns. Fewer than 30% reported regular assessment of financial and insurance concerns. Once identified, most concerns were adequately managed by care teams, the exception being practical concerns. Interviewees highlighted quality survivorship care in general with some internal and external barriers. Participants recommended increasing awareness of available services and improving care coordination and navigation related to survivorship care.

Conclusions Results suggest that while survivorship care may align with the national standards related to physical and emotional concerns, there is critical need to improve assessment and management of financial concerns.

Implications for Cancer Survivors Cancer centers should conduct comprehensive screening that includes financial and other practical concerns and implement processes for connecting survivors with needed services.

Keywords Cancer · Cancer survivorship · Quality improvement · Standard of care

Introduction

In 2025, 18.6 million people living in the United States (US) are survivors of cancer [1]. By 2040, this number may reach 26 million [2]. Roughly two million cancer diagnoses are made in the US annually [3, 4] with overall 5- and 10-year survival rates at 70% and 48%, respectively [2]. Consequently, there is a rapidly increasing need in cancer care to attend to the physical, psychological, social, and functional concerns of survivors [5] while addressing financial barriers to healthcare utilization [6].

Following the 2006 Institute of Medicine report, *From Cancer Patient to Cancer Survivor: Lost in Transition* [7], there have been quality improvement efforts around survivorship care (SC). Clinical practice guidelines and

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For National Standards for Cancer Survivorship Care – Special Section

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accreditation standards routinely require SC plans [8], with the American Society of Clinical Oncology and Multinational Association for Supportive Care in Cancer emphasizing care planning in person-centered care [9]. Additionally, proactive, personalized, and comprehensive SC across the cancer care continuum is highlighted by the National Cancer Institute (NCI) [5], American Cancer Society [10], National Comprehensive Cancer Network [11], and Commission on Cancer (CoC) [12].

While important, implementation of evidence-based SC as part of routine cancer care faces barriers. Nationwide studies have found that standalone services for mental health, nutrition, and/or physical activity are frequently available to survivors [13]. However, specialized SC clinics are uncommon with only 31% of CoC accredited facilities and 41% of NCI Community Oncology Research Program institutions reporting dedicated SC clinics [13, 14]. Within cancer centers, SC is often delivered by extant cancer treatment teams as opposed to dedicated, multidisciplinary teams or well-established, integrated networks of specialty providers. This is due to few providers with survivorship specialization, as well as limited awareness of, or referrals to, survivorship programs [14]. Gaps also exist in understanding best practices for implementing systematic surveillance and care navigation for cancer's physical and psychosocial late effects and common issues like financial toxicity [15].

To improve SC implementation, the 2024 NCI National Standards for Cancer SC define standards and actionable recommendations for health systems [5]. These standards outline components of health system policy, processes, and evaluation/assessment central to quality SC [5]. As these new standards renew a focus on SC, this study relies on survivor-reported experiences to 1) describe the degree to which SC at one NCI Designated Comprehensive Cancer Center aligns with standards pertinent to health system processes [5] and 2) identify barriers to and facilitators of accessing SC.

Methods

Study design

This convergent mixed-methods study employs standardized surveys and semi-structured interviews. The setting is a university-based NCI Designated Comprehensive Cancer Center in the southeast US.

Participants and procedures

Survivors were recruited from four outpatient clinics for solid tumors. Inclusion criteria were: 1) at least 18 years old, 2) at least three months into active treatment or being followed post-treatment, and 3) able to understand English well enough to comprehend and answer eligibility and other study questions.

After researchers pre-screened for eligibility using the electronic health record (EHR), potentially eligible survivors were approached in clinic, informed about the study, screened for eligibility, and asked to provide verbal consent to participate. Recruitment went clinic-by-clinic, with attention to balanced representation of survivors with respect to treatment phase (active vs. post-treatment), disease stage (early vs. late), and residence (rural vs. nonrural). Participants were asked to complete a 30-min survey either on paper or online. If participants did not finish in the clinic, they could do so at home. Participants (N = 150) received a $50 gift card for survey completion, with surveys done between February and August 2025. Additionally, some participants were purposively selected based on criterion and invited to complete a semi-structured interview, the goal being to amass representative perspective with in-depth exploration. Select participants were invited after completing their surveys in clinic to gain a roughly even distribution of cancer sites, active vs. post-treatment, rural vs. nonrural residence, and genders in the interview subsample and recruitment stopped once each perspective was adequately represented. Eighty participants were invited and 17 completed interviews. Interviews (mean [SD] minutes 24.2 [7.6]) were audio recorded using HIPAA-compliant Zoom software and completed between April and August 2025. Participants received a $50 gift card or check for interview completion. Recordings were transcribed verbatim and cleaned using Rev software. The university's Institutional Review Board approved all procedures (IRB #98,262).

Measures

EHR data

Clinical information about cancer site, stage, and treatment was extracted from the EHR as was county of residence, which determined rural versus nonrural status (2023 USDA Rural–Urban Continuum Codes 4–9 = rural) [16].

Survey

Sociodemographic items collected data on age, race and ethnicity, gender, relationship status, education, household income, and employment status using questions and response options similar to those in US population surveys [17]. To assess how the national standards were addressed across key domains (physical, emotional, practical, social, financial, and insurance concerns), we followed an approach similar to the Patient-Centered SC Index regarding clinicians' assessment of and listening to concerns [18] (see

Supplemental File 1 for the survey). For each domain, participants were asked to report how often their cancer treatment team assessed the domain using a 5-point Likert scale from "has asked me at every visit" to "has not asked me". Like an item from the Patient-Centered SC Index that asked, "Did your clinicians listen carefully to concerns related to…" participants were asked how often their treatment team "listened carefully" when they discussed concerns using a 4-point Likert scale from "not at all" to "completely," with "don't know" as another option. For any domain-specific concern experienced, participants were asked whether the treatment team "advised or helped manage" the concern (Yes/No) or referred them elsewhere (Yes/No). Finally, participants reported being asked about lifestyle behaviors (e.g., tobacco use, Yes/No) and whether they had received related advice or management.

Interview

Informed by the national standards, a semi-structured interview guide with a core question set and follow-up probes was utilized (see Supplemental File 2). Questions centered around participants' experience with care delivered at the cancer center (e.g., "If you had concerns, what recommendations or referrals were you given?").

Data analysis

EHR and survey data

Descriptive statistics characterized participants' sociodemographic and clinical characteristics. The standards state survivors should be "assessed" for various concerns (e.g., practical and social) and "provided with" or referred for applicable resources, support, and services to manage any concerns [5]. This study examined alignment with the standards by measuring three aspects of care delivery across domains/concerns. First, survivors were considered to have been asked about a domain if they reported being asked at "most/every" visits to reflect the importance of regular assessment. Additionally, this approach reflects the standards' reference to survivors "being assessed at multiple points" for the physical and emotional/psychological effects of cancer, an expectation we extended to all concerns [5]. Second, concerns were listened to if participants reported their treatment team listened "mostly/completely" to them. Finally, concerns were managed if participants answered "yes" to receiving advice/help or referral about a concern they experienced. The proportion of participants who reported being asked, listened to, and having concerns managed was examined by domain. Chi-square tests or Fisher's exact tests (cell counts < 5) evaluated group differences (i.e., treatment phase, disease stage, and residence) related to participants' concerns and lifestyle behaviors. Pairwise deletion was used to manage missing data for each individual test. Statistical significance was set at $p < 0.05$. Analyses were conducted using IBM SPSS Statistics version 31.

Interview data

Interview data were analyzed using directed content analysis [19]. One coder (SBH) developed an initial codebook with four parent codes (SC, facilitators and barriers, suggestions and recommendations, and stories) plus child and grandchild codes, all with operational definitions and illustrative quotes. This codebook was reviewed (JLB, MAW) and subsequently revised. Next, SBH and MAW completed one round of consensus coding for three transcripts, with 94.8% agreement across codes, after which three additional transcripts were coded to ensure consensus was maintained. Coders then worked independently to code all remaining transcripts using NVivo version 15, with questions resolved via consensus meetings. Afterwards, MAW and JLB manually applied thematic analysis to code-specific reports from NVivo. At this step, coders worked independently and then met to compare themes and reach consensus about both the themes and illustrative quotes.

Results

Sample characteristics

In total, 206 eligible survivors were approached about the study, 179 survivors consented, and 150 survivors completed the survey. Table 1 details participants' sociodemographic and clinical characteristics (N = 150). Ages ranged from 22 to 83 years (mean [SD] age 59.03 [13.13]). Participants were predominantly White (90.67%) and non-Hispanic (98.67%). Most participants were female (62.00%). Roughly one-half lived in rural counties (53.33%). Educational backgrounds varied, such that while 21.33% only had a high school education, 42.67% earned a college degree. Annual household income was widely distributed, with 23.33% reporting incomes below $25,000 and 20.00% reporting incomes above $100,000. Regarding employment, half were either working (31.33%) or disabled (19.33%). The most common cancer sites were breast (28.00%), lung (16.00%), head/neck (12.00%), and gastrointestinal (19.33%). Forty-one percent of participants had early-stage disease and 50.67% were in active treatment.

Perceptions of SC processes

Survey results

Figure 1 displays the extent to which participants reported being asked about, listened to, and having concerns

Table 1 Sociodemographic and clinical characteristics of participants (N = 150)

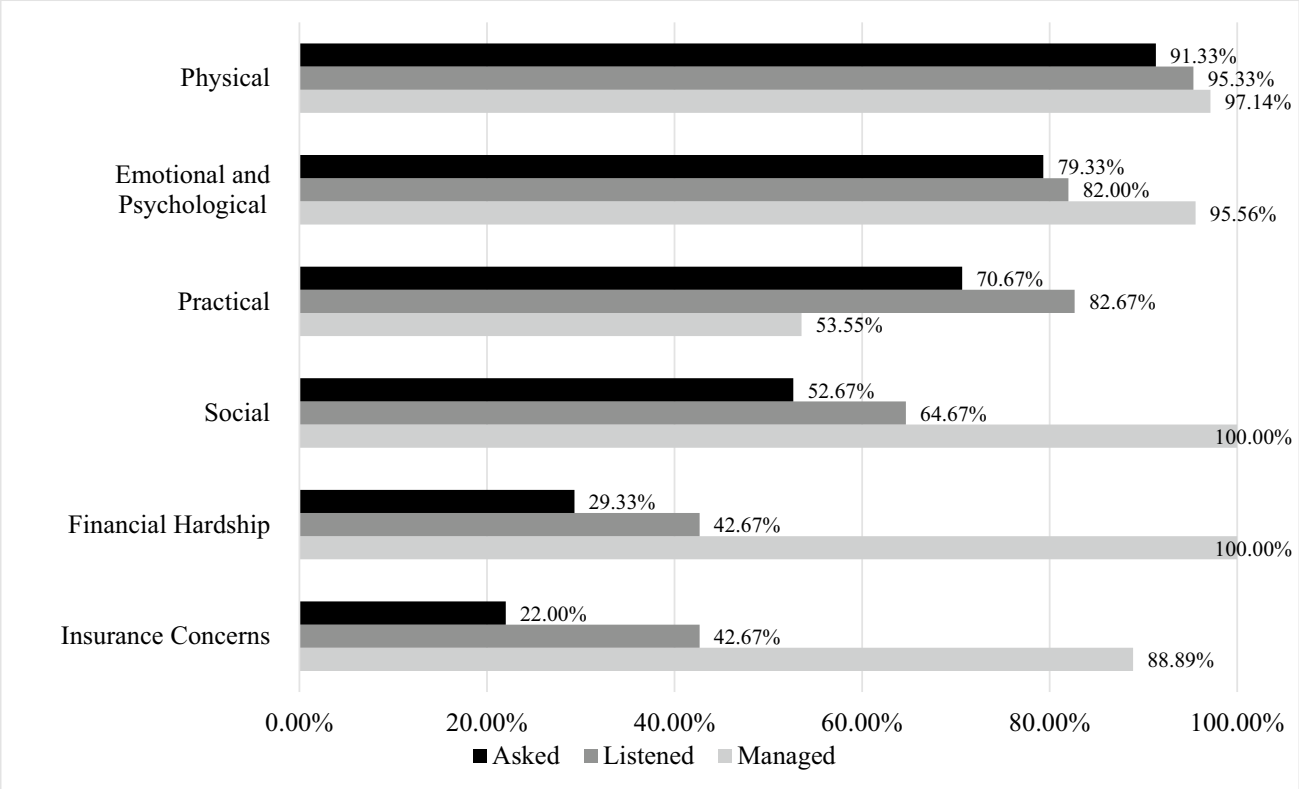
Variable	No. (%) of participants
Age in years	
Mean (SD)	59.03 (13.13)
Range	22–83
Race	
White	136 (90.67%)
Black	9 (6.00%)
Asian	1 (0.67%)
Middle Eastern/North African	1 (0.67%)
Missing	3 (2.00%)
Hispanic ethnicity	2 (1.33%)
Female gender	93 (62.00%)
Rural residence[a]	80 (53.33%)
Relationship status	
Married	92 (61.33%)
Separated/Divorced	27 (18.00%)
Widowed	16 (10.67%)
Single, never married	12 (8.00%)
Education	
Less than high school diploma	14 (9.33%)
High school diploma or GED	32 (21.33%)
Technical or vocational school	13 (8.67%)
Some college	27 (18.00%)
College graduate	45 (30.00%)
Post graduate degree	19 (12.67%)
Annual household income	
< $25,000	35 (23.33%)
$25,000–49,999	41 (27.33%)
$50,000–99,999	32 (21.33%)
≥ $100,000	30 (20.00%)
Missing	12 (8.00%)
Current employment status	
Working full- or part-time	47 (31.33%)
Homemaker	9 (6.00%)
Retired	57 (38.00%)
Disabled	29 (19.33%)
Unemployed	7 (4.67%)
Missing	1 (0.67%)
Cancer treatment phase	
Active treatment	76 (50.67%)
Post treatment	74 (49.33%)
Cancer type	
Brain	5 (3.33%)
Breast	42 (28.00%)
Gastrointestinal	29 (19.33%)
Genitourinary	10 (6.67%)
Gynecologic	8 (5.33%)
Head and neck	18 (12.00%)
Lung	24 (16.00%)
Skin	4 (2.67%)
Soft tissue and/or bone	9 (6.00%)
Unknown	1 (0.67%)
Cancer stage or grade[b]	
Early (0-II)	64 (42.67%)
Late (III-IV)	72 (48.00%)
Unknown	3 (2.00%)
Stage or grade not found in electronic health record	11 (7.33%)

[a] Rural residence was determined from participants' county of residence in the electronic health record and defined using 2023 USDA Rural–Urban Continuum Codes, with codes 4 to 9 designated as rural [16]

[b] Brain cancers are graded not staged

managed. Participants most frequently reported that treatment teams asked about physical concerns ($n = 137/148$, 91.33%), tobacco use ($n = 130/149$, 87.25%), physical activity ($n = 118/150$, 78.67%), diet ($n = 110/149$, 73.83%), and emotional concerns ($n = 119/147$, 79.33%), with financial hardship ($n = 44/148$, 29.33%) and insurance ($n = 33/147$, 22.00%) assessed the least. This pattern was similar to reports related to treatment teams listening to participants' concerns. Management of concerns—defined as providing advice, help, or referrals—was reportedly high across domains among those who reported experiencing each concern and communicating with their care team about it, particularly for physical ($n = 102/105$, 97.14%), emotional ($n = 44/46$, 95.56%), financial/insurance ($n = 21/21$, 100%), and social ($n = 22/22$, 100%) concerns. Management of lifestyle behaviors was less prevalent, but still moderate-to-high (e.g., diet: $n = 103/110$, 93.64%, tobacco use: $n = 91/130$, 70.00%).

With one exception, there were no significant group differences in participants being asked, listened to, or managed when analyzed by treatment phase (active vs. post-treatment) and cancer stage (early vs. late stage) (data not shown). The exception was a large difference in management of practical concerns (e.g., transportation), with 96.00% of participants in active treatment receiving management compared to none of the post-treatment survivors ($p < 0.001$). Comparisons by residence revealed some significant group differences (data not shown). Participants from rural compared to non-rural counties reported being less likely to be assessed regarding physical and financial concerns (88.61% vs. 97.10%; $p = 0.049$ and 22.78% vs. 37.68%; $p = 0.048$, respectively), and less likely to report being listened to about practical and social concerns (76.92% vs. 94.12%; $p = 0.004$ and 58.44% vs. 78.79%; $p = 0.009$, respectively). Regarding lifestyle behaviors, rural participants reported being less likely to be asked about diet (65.82% vs. 82.86%, $p = 0.018$), physical



[a] *"Asked" indicates whether the provider inquired about the concern (yes/no). "Listened" indicates whether participants perceived that their provider adequately listened to the concern (yes/no). "Managed" was assessed only among participants who reported experiencing a concern in the domain and coded as "yes" if the provider either advised or otherwise helped manage the concern or referred them to a specialist or service for assistance.*

Fig. 1 Participant Report of Assessment and Management of Concerns Across Survivorship Care Domains by the Care Team[a]

activity (72.50% vs. 85.71%, $p = 0.049$), and alcohol use (43.64% vs. 63.79%; $p = 0.032$).

Interview findings

Table 2 shows the codes applied to the interview data, with illustrative quotes. Below is a summary of themes identified.

Participants often commented on the generic nature of assessment wherein they were frequently asked questions such as "Is there anything you need?". Structured, focused assessment beyond routine check-in questions about pain, depression, tobacco use, and medical history changes were less common (see Table 2, A). Participants noted that symptom, function, and wellbeing screenings were often conducted ahead of appointments via the patient-facing portion of their EHR or in clinic by rooming staff or nurses, with inconsistent follow-up assessment by oncologists (Table 2, B). When concerns or problems were identified, though, participants felt that providers—often described as functioning as a "team"—adequately addressed them. Referrals to social workers, nutritionists, genetic counselors, and physical therapists were common, with other referrals like dentistry recommended only for certain disease sites (e.g., head/neck). Participants rarely recalled referrals for services to palliative care and fertility specialists. Notably, participants expressed low awareness of the cancer center having established "SC Visits", with few interview participants reporting an offer to meet with a dedicated SC nurse after treatment completion (Table 2, C-E). Finally, participants described the experience of shared decision-making and involvement in care planning, and believed their preferences and priorities were respected.

Internal barriers to and facilitators for care were defined as factors within the health system that either undermined or contributed to delivery of SC, respectively. Common internal barriers included suboptimal care coordination among multidisciplinary teams, difficulty accessing supportive care services, logistical challenges in navigating the physical hospital/clinic environment, and poor provider communication or insensitivity (Table 2, F–H). Common internal facilitators included effective communication with providers who were approachable and accessible, providers who encouraged shared decision-making, providers who meaningfully engaged in informed consent to treatment and follow-up, coordinated multidisciplinary care teams who delivered

Table 2 Illustrative quotes by code and relevant national standard(s) for cancer survivorship care

Code	Reference	Illustrative Quote	Survivorship Care Standard
Processes[a]	A	"They just asked me how I'm feeling. I mean, if I'm having any pain or if I'm depressed, stuff like that because I do, if I'm having a good day or bad day, sometimes I have good days, I have bad days." (170, active treatment, late-stage unknown site, nonrural)	**Health system processes:** Assessed at multiple timepoints for: physical effects; emotional and psychological effects
	B	"Well, the nurses always ask you about pain that have their set schedule of things they ask you. The providers are probably a little more vague. They ask more just how you're doing. And so if you have something, you bring it up." (204, post treatment, early-stage head & neck, rural)	
	C	"They ask if [I'm] in any pain that if there's anything new, any kind of symptoms or anything that I'm going through at that time that I have any kind of concerns or which if I have any symptoms, then they will talk about whatever the symptoms is. Some of the chemo was coughing, rash and stuff, so they would talk about that and what to treat or different, whatever symptom I'm having, they will address that symptom." (402, post treatment, early-stage breast, rural)	
	D	Interviewer: "At any point in your treatment so far at Markey, has anyone asked if you did want to meet with the survivorship care nurse?" Interviewee: "I don't recall." (306, active treatment, late-stage head & neck, rural)	**Health system processes:** Provided with access and referral to a survivorship program that addresses the needs of cancer survivors either on-site, through telehealth, or by referral
	E	Interviewer: "And then at any point in your treatment at Markey, has anyone asked or did anyone ask if you wanted to meet with a survivorship care nurse directly?" Interviewee: "No, I didn't know that it existed until you started talking to me about it." (469, post treatment, late-stage soft tissue/bone, rural)	
Internal Barriers[b]	F	"I don't think I've brought it up that I think there's hard dealing with all the leftover part. And I think this is the way it feels, is that they're feeling very good because I'm in remission and my tumor is gone. And so for them it feels like it's a win. The issues I have with eating and it's very hard to gain weight… And the challenges with eating are every day, you eat every day, so you don't get a day off. And that's really a struggle. And they don't seem empathetic about that… so they're doing what they have in their tool bag, but it doesn't feel like there's a recognition of how hard that is to deal with it." (204, post treatment, early-stage head & neck, rural)	**Health system processes:** Assessed at multiple points in their follow-up care for physical effects during and following cancer treatment, including monitoring for late effects and chronic conditions, and provided with treatment and/or referrals
	G	"What I have trouble with is that I had so many different doctors and there doesn't seem to be anyone in particular that looks at the overall picture. So I had surgery, I had chemo, I had radiation, and I got my regular doctor involved in this. And I'm still having the stamina issues and it's like I need one doctor to say this is what we need to do, communicate with the other doctors, me trying to talk to each one of 'em… I don't know who to ask the right questions to sometimes…" (469, post treatment, late-stage soft tissue/bone, rural)	**Health system processes:** Engaged in the care planning process including discussion of shared goals of care, advanced care planning, and coordination of care with providers and services
	H	"I went to [another cancer center] first before I came to you guys. And I feel like they reach out a lot more, I guess their social workers and stuff and people they still call me and stuff to check in where I'm not there. But I guess maybe that, I guess coming to the patient with the resources more so than them having to find them and come to them kind of thing." (376, active treatment, early-stage breast, rural)	**Health system policy:** Provision of information about support services for survivors based on their needs

Table 2 (continued)

Code	Reference	Illustrative Quote	Survivorship Care Standard
Internal Facilitators[b]	I	"It's like they don't make a decision unless you're 100% know what they're talking about. They take the time. Every one of the doctors that I've seen there has actually took the time to explain to me what's going on. And they don't use big doctor terms because I don't understand them. And so they make sure that I understand. They send me notes after my visits so I can read it because sometimes you just get overwhelmed and you don't remember. They give me my options, the medicine what it can and cannot do, what it might not do. They don't give you false hope and they don't, they're just straightforward, but in a compassionate way." (226, active treatment, late-stage gastrointestinal, rural)	**Health system processes:** Engaged in the care planning process including discussion of shared goals of care, advanced care planning, and coordination of care with providers and services
	J	"I mean, I really like the services that they provide and I really think that every time I go in, the people are so friendly as far as [cancer center], I really think they really try to, it's not a number, it's the way I feel… I'm not a number. They actually care about what's going through." (402, post treatment, early-stage breast, rural)	-
	K	"The nurses that was compassionate, the doctors, the surgeons, I have no issues with none of the oncologist team and kudos to the nurses… it's like family, you're here every week. So you see 'em every week. And they would tell me what I was going to be expecting." (369, active treatment, early-stage breast, nonrural)	-
External Barriers[c]	L	"I smoke and he [provider] jumped all over me about, I'm trying to quit smoking, and he had ordered me patches and stuff and I can't afford, it's like 75 bucks. I can't afford patches over the lozenges, so I called up and left a message for 'em, said if they could help me find somebody that can help me cover it, my card won't cover it. So I'm stuck here trying to figure out to get help with do that." (170, active treatment, late-stage unknown site, nonrural)	**Health system processes:** Assessed for lifestyle behaviors and provided with recommended strategies for management and appropriate referrals or education as needed; Assessed for financial hardship/toxicity and concerns regarding insurance coverage, and provided with resources and support as needed
	M	Interviewee: "I know I talked to 'em [social work] when I found out I was doing radiation and stuff because they were talking about the services they offer. I think it's someplace you can stay and they had told me about that." Interviewer: "And did you end up taking them up on the housing services? Interviewee: "I didn't. Just because I have three little ones and I can't be away from 'em that long." (376, active treatment, early-stage breast, rural)	**Health system processes:** Assessed for financial hardship/toxicity and concerns regarding insurance coverage, and provided with resources and support as needed
	N	"I need monies to get my car fixed or I need another car. My car is down. I've been using my girlfriend's car for about a year then… I've been using her car before she take me, she's my caregiver." (487, post treatment, late-stage genitourinary, rural)	
External Facilitators[c]	O	"Originally after the surgery, we talked about FMLA and having the forms filled out. And I do have [insurance type], so there really aren't, thank God, any financial concerns regarding my treatment. And other than that same thing, because I was able to go back to work and I had FMLA coverage, there weren't any financial concerns." (370, active treatment, early-stage breast, nonrural)	-
	P	"Let's see, I think it was like, is it Komen or something? They helped to pay one of my core payments because work had really I worked during chemo, but the social workers were really helpful with getting all that filled out and doing all the things." (376, active treatment, early-stage breast, rural)	-
	Q	"The last time I was in the hospital, I ended up with two blockages and they ended up, because I get dehydrated really easy now, so they've actually stepped up to go ahead and let me get fluids at my local hospital twice a week." (226, active treatment, late-stage gastrointestinal, rural)	-

Table 2 (continued)

Code	Reference	Illustrative Quote	Survivorship Care Standard
Suggestions and Recommendations[d]	R	"Basically, I needed somebody to give me more information and there were certain doctors that would tell me stuff. And actually it was very nice to have it in [patient-facing EHR] all the information there because I mean certain doctors, I could read all that information and verify and talk with them and know I understood it enough that it was worthwhile for me to read all that. And in some cases I could correct them if I saw things that were wrong in there. I wish all the doctors would put that in there and have access to all my information that way. I understand not everybody would understand everything that was there, but I mean, what I learned was very helpful." (469, post treatment, late-stage soft tissue/bone, rural)	**Health system processes:** Engaged in the care planning process including discussion of shared goals of care, advanced care planning, and coordination of care with providers and services
	S	"It's not that I want them to start having me see a counselor. I could do that on my own too… But I think that the talking part, if they could just recognize that, gosh, this really sucks your life turned on its head and I feel like I'm about half of the person I used to be. So having some more of that kind of compassion as part of the treatment or just being willing to stay with that when somebody brings up about, it's really hard to eat, they don't have to do a lot, but they could just stay with that for a minute and just, yeah, I'm sure it's hard even just acknowledging it, acknowledging the emotional part of it rather than just, oh yeah, people have xerostomia after treatment." (204, post treatment, early-stage head & neck, rural)	-
	T	"This might sound silly but just being up there I think that some kind of services to feed the family that's there, that sounds like so silly, but with my family there, we've had to stay there up to seven days and just the cost of them eating was over $600. And I mean that blew my mind because everything in the cafeteria was so high and I don't know, it's like maybe if they could get some kind of a voucher to eat down there, that would be great because in all reality, when you got somebody say dying of cancer, you don't want to worry about having to pay for something to eat. And it's hard not only for the family but for the patient too because they're worried about their family eating, but they don't want to be alone." (226, active treatment, late-stage gastrointestinal, rural)	-

[a]Processes: Statements made about cancer center organizational processes regarding survivorship care (e.g., access provided, referral processes, follow-up care assessments, lifestyle and/or behavioral assessments, care planning engagement, etc.)

[b]Internal barriers and facilitators: Barriers or facilitators to survivorship care that can be directly connected to the cancer center or within the health system

[c]External barriers and facilitators: Barriers or facilitators to survivorship care that are not connected to or are beyond the direct control of the cancer center or health system

[d]Suggestions and recommendations: Statements made about explicit suggestions and recommendations regarding survivorship and/or survivorship care

personalized care, and seamless integration of supportive care into overall cancer care (Table 2, I-K).

External barriers and facilitators to care were defined as factors outside the control of the health system that nonetheless strongly impact survivors' healthcare access, quality, and satisfaction. Common external barriers included insurance coverage limitations or absence, limited financial means, practical problems related to travel distance, and suboptimal care coordination between the cancer center and community-based health services like primary care (Table 2, L-N). External facilitators included insurance coverage of comprehensive services, partnerships and coordination between the cancer center and community-based care health services, and assistance from foundations and other nonprofit organizations to help with financial expenses and tangible resources like wigs (Table 2, O-Q).

Finally, participants offered suggestions to improve SC. Themes centered around the cancer center providing more information about treatment and available services, more direct offers of assistance and referral so survivors have less obligation to advocate for their needs or find available solutions, and additional help for survivors in coping with the stress of cancer care through clearer communication, more compassion, and family support (Table 2, R-T). Despite suggestions for quality improvement, participants tended to express high satisfaction with the care they received due to the perception that they underwent fully informed consent with questions answered and concerns addressed. Additionally, high satisfaction was tied to appreciation for personalized treatment approaches, with one participant noting, "I'm not a number. They actually care…" (SID 402).

Discussion

This mixed-methods study assessed the quality of SC experienced by survivors at an NCI Designated Comprehensive Cancer Center using the National Standards for Cancer SC as a guide. This is one of the first studies to directly examine clinical practice alignment with the national standards from the perspective of survivors. Results suggest that to meet the national standards, health systems will likely need to adopt systematic assessment and comprehensive management of financial concerns, including cost of care and health insurance concerns, as well as practical concerns (e.g., returning to work). Health systems will also need to attend closely to the needs of vulnerable populations who may shoulder disproportionate burden of these concerns and experience unique barriers addressing them, such as rural survivors.

The national standards state that survivors should be assessed at "multiple points" in their follow-up care for the physical and emotional effects of cancer, as well as assessed for practical effects, lifestyle behaviors, and financial hardship, and subsequently provided with appropriate treatment, resources, or referrals [20]. The present study demonstrates that physical effects were the most assessed, listened to, and managed domain, followed closely by emotional concerns. No differences in these processes were observed between survivors in active vs. post-treatment and with early vs. late-stage cancers. This positive finding is unsurprising given the prominence of physical symptom management in cancer care and the prioritization of screening and treatment for psychological concerns in the National Comprehensive Cancer Network Guidelines for Distress Management [21]. Still, survivors in this study emphasized the importance of effective, compassionate communication from providers and care coordination as facilitators of quality SC. Recommendations for improving care included increasing direct, proactive offers of assistance and referral to reduce the need for self-advocacy and offering additional help for survivors to manage the stress of treatment through improvements in communication and increased material support for caregivers.

In contrast to physical and emotional concerns, survey data on survivor reports of assessment for financial concerns suggested a major practice gap: less than one-third reported being assessed for financial or health insurance concerns. Interviews with survivors complement this survey finding, as they demonstrated the salience of practical barriers to SC related to cost. Among participants who reported experiencing a financial or insurance concern and were able to communicate the concern to their care team, 100% reported that their concern was managed. While this finding is encouraging, demonstrating the capability to assist survivors when their needs are identified effectively, it could be an artifact of a small sample size and there is yet room for improvement in assessment. That said, the low assessment rate found in this sample deviates from that in some other studies. In national studies of NCORP and NCCN sites, 72–78% of sites reported having a financial screening process in place for patients [22–24]. At the survivor level, however, the occurrence of financial screening may not be implemented optimally, with 58% of breast cancer survivors in one sample reporting that they were not asked about financial stressors by their care team [25]. Thus, it is likely that even when there exists a policy that mandates screening for financial hardship and insurance concerns, and a process in place for population-level screening, there is still room for improvement regarding the reach of that screening/assessment.

Inclusion of financial and health insurance concern assessment and management in the standards recognizes the public health crisis surrounding cost of care and financial toxicity in cancer. Financial toxicity, or hardship due to medical care costs [26], affects 28–48% of survivors [27], with evidence that worse financial toxicity is related to active treatment [28], late-stage diagnosis [28], and time

since cancer diagnosis [29]. While differences in assessment of financial concerns by cancer treatment status were not observed here, differences by rural vs. non-rural residence were observed, indicating that financial toxicity may be more likely to go unnoticed among survivors living in rural areas. It's unclear why, but this may be due to care teams assuming that rural survivors do not want to discuss financial and other personal matters publicly due to cultural norms, an assumption that may inadvertently lead to inadequate assessment of financial (and practical) concerns. Financial toxicity is closely tied to health insurance concerns, wherein inadequate coverage, high treatment costs, and impacts to employment exacerbate toxicity [30–32]. The field has been grappling with how to address financial toxicity in cancer care for years, particularly in light of therapeutic advances conferring longer survival and treatment courses for likely incurable cancers [2]. While optimal methods for screening and managing financial toxicity remain to be determined, multi-level solutions are needed, including improved rates of proactive screening, access to financial navigation, and empathetic provider communication about financial hardship [22, 33–35]. Cancer centers need to implement proactive, standardized assessment of financial/insurance concerns while recognizing that assessment alone is insufficient and the burden of communication and management should not fall on survivors. Financial toxicity is an area of growing attention given projected costs of cancer care and potential for increased premiums and increased rates of under and uninsured survivors [36], so focused attention to this component of the standards is critical.

Overall, findings emphasize the importance of population-level, systematic, prospective, and comprehensive assessment of the needs – and lifestyle behaviors – of both active treatment and post-treatment survivors. In this study, which focuses on survivors' reports of cancer care teams' practices, it was found that cancer treatment teams effectively managed survivors' concerns once they were made aware, but social and financial/insurance concerns (and to a lesser extent practical concerns) often required survivors to seek assistance. Given that social determinants and practical matters like insurance coverage, travel distances, and financial hardship can delay or otherwise disrupt cancer care delivery and negatively influence cancer outcomes, it is imperative that the onus not be on survivors to voice concerns in these domains, but instead for preventive measures to be adopted at the system-level. Cancer centers should ensure that assessment processes are standardized for each of the health system processes identified in the SC standards and that providers are knowledgeable about resources or referral options to manage each concern at the point of care to reduce possible disruptions in treatment. Moreover, periodic audits of assessment processes can be a valuable tool to ensure that screening tools are working as planned and that survivors' needs are being met. Gathering information directly from survivors' can provide valuable insight in identifying gaps between policies and processes in action and inform efforts to improve on-the-ground screening practices.

Toward that end, the national standards provide a critical first step in a roadmap for improving SC quality in the US. Assessing direct alignment with the national standards was difficult in the present study as each standard provides a general description of important processes but little direction for measurement. Cancer care delivery researchers and leaders can move implementation forward by operationalizing guidelines to include clearly defined steps for management of concerns and benchmarks for concordance, as well as developing a uniform process for cancer centers to independently assess/rate themselves. Prior research using EHR review or de-identified patient data to assess overall cancer care quality [37] and adherence to Institute of Medicine practice guidelines [7] for SC plans [38–41] offer a blueprint for evaluating SC delivery.

The foremost study limitation is that this study did not measure every process indicator included in the standards comprehensively. For example, we measured care of survivors' overall "physical" concerns as opposed to separating out late-effects and chronic conditions. Second, the survey/interview data were self-reported, meaning the results may be impacted by recall and/or social desirability bias. Relatedly, this study's assessment of alignment with the standards relies on survivor report not EHR or claims data, or other objective indicators, which have value. Third, while it met the study aims, the survey used face valid items as opposed to psychometrically validated instruments. Fourth, as data were collected from one cancer center, the results likely do not generalize—without qualification—to cancer centers nationwide. Finally, the subset of participants who were interviewed was small, so qualitative data should not be viewed unequivocally as representative of the population, but as a means of expanding upon survey results. Even with these limitations, this study provides a robust, novel clinical practice assessment of SC quality, guided by the new national standards. Future implementation research should help refine the national standards to ensure they are both practical and impactful, particularly regarding processes and evaluation and cancer centers should employ empirically supported quality improvement methods like Plan-Do-Study-Act [42] to implement the national standards (possibly with special focus on financial matters) with the important goal of improving access to quality care for all survivors.

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Data availability Survey data are available upon reasonable request to the corresponding author. The qualitative data underlying this article cannot be shared publicly due to the desire to protect the privacy of individuals who participated in the study. Deidentified interview transcripts will be shared upon reasonable request to the corresponding author.

Declarations

Ethics approval This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the University of Kentucky Institutional Review Board (IRB #98262).

Consent to participate Informed consent was obtained from all individual participants included in the study.

Competing interests The authors have no relevant financial or non-financial interests to disclose.

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